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Alternative splicing and intron retention: Their profiles and roles in cutaneous fibrosis of systemic sclerosis

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ABSTRACT

Background: Alternative splicing (AS) and intron retention (IR) implicated in multiple pathophysiological processes, have rarely been reported in systemic sclerosis (SSc).
Methods: We integrated bulk RNA-seq and 4D label-free mass spectrometry to perform a multi-omics analysis of AS and IR in SSc skin tissue and fibroblasts. RMATS and iREAD were used to identify AS and IR, which were validated by real-time PCR. Spearman correlation and the LASSO method were employed to assess correlations among clinical features, introns, splicing factors (regulators of AS) and proteins.
Findings: AS profiles showed distinct alterations in SSc skin tissue, with the most pronounced changes occurring in IR. AS and IR were associated with total modified Rodnan skin score (mRSS) and local skin score. Upon TGF-β stimulation, fibroblasts exhibited significant alterations in IR profiles, affecting genes related to fibroblast proliferation and collagen fibril organization. A comprehensive integrated analysis of introns, exons, and proteome profiles revealed that IR exerted a negative impact on protein expression, with certain changes being under intronic control. RT-PCR confirmed the presence of intron and exon-derived sequences of *CTTN, OGA, MED16* and *PHYKPL*. Additionally, notable changes were observed in the regulatory network of splicing factors in SSc skin tissues. These factors are also involved in fibrosis pathways and correlated with clinical features.
Conclusion: Totally, abnormal AS, IR profiles and splicing factors were identified in SSc, altered IRs and splicing factors participated in fibrosis-related pathways. IR exerted a negative impact on protein expression in TGF-β-stimulated fibroblasts. Clarification of the IR mechanisms will provide new insights into the pathophysiology of SSc.

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1. Introduction

Systemic sclerosis (SSc) is a heterogeneous autoimmune disease that involve multiple systems with skin thickening being the most common clinical feature [1,2]. Fibroblast activation is at the center of the

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pathogenic triad of fibrosis, immune dysfunction and microvascular damage [1,2], which leads to excessive synthesis of extracellular matrix component [3]. Although several drugs treating pulmonary fibrosis have been applied in clinical management, their efficacy are not satisfactory and drugs targeted to fibrosis are still urgent, which needs the exact clarification of fibrosis pathogenesis [4].

It was shown that mRNA abundances contribute to only 40 % of variation in protein abundances [5]. The expression level of proteins is dynamically regulated by various cellular processes, such as alternative splicing (AS). AS is widespread in cells and generates multiple transcripts and protein isoforms, consisting of skipping exons (SE), alternative 3′ splicing site (A3SS), alternative 5' splicing site (A5SS), intron retention (IR) and mutually exclusive exons (MXE) [6]. IR, which has attracted increasing attention recently, refers to the process in which specific introns are retained in mature mRNA instead of being spliced out and is regulated by a complex splicing machinery involving spliceosomes and upstream regulatory factors, which is so intricate that it involves nearly 300 genes [7].

Introns are preferentially conserved across organisms and species [8]. IR can provide materials for evolution and serve various functions within cells, including introducing complexity to transcripts in mature cells and protecting against starvation stress, and IR can be decreased in response to certain stimuli [8–14]. IR increases during granulocyte differentiation and terminal erythropoiesis to remold the nucleus structure and decreased during CD4$^+$ T-cells activation [14–16]. Transcripts containing IR (IRTs) can be localized from the dendrite to the cell soma in neurons thereby enabling their splicing and translation as soon as a stimulus is received [13].

For a long time, despite nearly 95 % of human genes undergoing alternative splicing, including intron retention (IR), the latter has been consistently underestimated [17]. In some cases, exon annotation can be transcript-specific and difficult to predict, especially when dealing with short exons [18]. Thus, in our previous studies, we developed algorithms called GTFtools and iREAD with the aim of identifying specific introns [19,20], which have been widely employed by researchers and applied in numerous studies pertaining to IR for pinpointing intron-retained regions of genes [21,22]. Additionally, we discovered altered IR profiles in the brains of individuals with Alzheimer's disease [23] and Idiopathic inflammatory myopathies [24]. IRTs have the potential to be translated into harmful truncated proteins, such as various isoforms of the tau protein associated with Alzheimer's disease [25,26]. Specific IRs were assessed being highly related to classic clinical features (such as Gottron's papules), muscle histopathological characteristics and laboratory examinations (such as inflammation cytokines, creatine kinase and antibodies) in Idiopathic inflammatory myopathies [24]. Numerous studies have been dedicated to seeking and developing drugs that target splicing factors and nonsense-mediated decay (NMD) (a common destination of most IRTs) [27–31].

However, up to this point, AS and IR have not been reported in SSc. Hence, our objective was to investigate the changes of AS and IR in SSc fibrosis progress. Additionally, we aimed to delve into the role of IR and potential mechanisms in the pathogenesis of SSc, while also identifying the upstream regulators of IR in SSc.

2. Methods

2.1. Cell culture

Skin fragments from sterile punches of normal individuals were cut into scrapes using sterile scissors and placed at the bottom of a flask in an incubator for 8h before being turned over. The cell culture medium was changed every 3 days and cell monolayers formed after 2–3 weeks. Cells were digested with 0.25 % trypsin liquid at 37 °C for 5–8 min and subsequently centrifuged at 2000 rpm for 5 min. Primary fibroblasts were purified after being passaged 2–3 times and identified by vimentin quantification. These primary skin fibroblasts were maintained in Dulbecco's modified Eagle's medium (DMEM; Gibco Invitrogen) supplemented with 10 % heat-inactivated fetal bovine serum (FBS). After 24 h in DMEM with 0.5 % fetal calf serum, fibroblasts at passage 6 were stimulated with TGF-β (10 ng/ml) (PeproTech) for 12 h. Additionally, the primary skin fibroblasts at passage 6 were pre-treated by the JNK inhibitor SP600125 (125 mM, 4 h) (Sigma-Aldrich, Steinheim, Germany), the ROCK inhibitor Y27632 (1 μM, 1 h) (Sigma-Aldrich), the TGFβRI kinase inhibitor SD208 (3 μM, 4 h) (Tocris, Nordenstadt, Germany), alpha V-containing integrins (*ITGAV*) inhibitor CWHM12 (1 μM, 1 h) (BOC Sciences, Shirley, NY, USA), or actin polymerization (*Actin*) inhibitor Cytochalasin D (10 μM, 1 h) (Abcr, Karlsruhe, Germany), and then incubated with WNT5a (100 ng/ml) (R&D Systems, Wiesbaden-Nordenstadt, Germany) for 48 h.

2.2. RNA isolation and quantitative real-time PCR

Total RNA was extracted using TRIzol (AG RNAex Pro reagent, ACCURATE BIOTECHNOLOGY, AG21102), following the manufacturer's instructions. cDNA was prepared using an Evo M-MLV RT Mix Kit (ACCURATE BIOTECHNOLOGY, AG11728), and real-time PCR (RT−PCR) was performed using a SYBR Green Premix Pro Taq HS qPCR kit (ACCURATE BIOTECHNOLOGY, AG11702) on an RT−PCR system (Applied Biosystem). The primers used to amplify the introns were designed specifically for this study and spanned the junction of introns and exons (Supplementary Table 1).

2.3. Bulk RNA-seq and data analysis

Total skin RNA samples (primary skin fibroblasts, stimulated by TGF-β (4 controls and 4 TGF-β stimulated samples) at passage 6) were selected for library preparation using the NEBNext Ultra RNA Library Prep Kit for Illumina (NEB, USA). The libraries were sequenced on Illumina platforms with the PE150 strategy at RiboBio Co. Ltd (RiboBio, China). Fibroblasts treated with WNT5a/JNK-i/ROCK-i/Actin-i/TGFBR1-i/ITGAV-i (3 controls and 3 samples in each group) were sequenced on an Illumina NovaSeq platform with a PE150 strategy at Novogene Co., Ltd. (Cambridge, United Kingdom).

The raw sequencing data (Fastq files) of primary skin fibroblasts, skin tissues (58 SSc patients and 33 control from the GEO database (GSE130955) [32]) and fibroblasts exposed to LiCl (GSE254716), JQ1 (GSE186961), TAK1 inhibitor (GSE232435), WNT5a/JNK-i/ROCK-i/Actin-i/TGFBR1-i/ITGAV-i (GSE222916) were aligned to the reference genome transcriptome (GRh38) using STAR v2.7(https://github.com/alexdobin/STAR). RMATS v4.2.0 [33] was employed to calculate AS events and identify significantly different AS events with false discovery rate (FDR) less than 0.05, which calculated the mean inclusion level (Inclevel) of a AS events in a group, calculated difference between 2 groups by employing likelihood-ratio test, and corrected P value using Benjamini-Hochberg adjustment. In each sample, AS events satisfied the threshold (0.05< inclusion level (Inclevel) < 0.95) (inclusion level indicates the ratio of AS occurrence in an isoform) were included to explore correlation with clinical indices and ECM score. Introns were counted by our previously published iREAD v0.8.9 [20], which used GTFtools [19] to identify independent introns without overlaying sequences with exons of other genes, and exons were counted by FeatureCounts v2.0.3 (https://github.com/ShiLab-Bioinformatics/subread). The data underwent differential analysis using edgeR v3.42.4 [34] and DEseq2 v1.40.2(https://bioconductor.org/packages/DESeq2). Gene Ontology enrichment analysis was performed using clusterProfiler v4.8.1(https://github.com/YuLab-SMU/clusterProfiler).

2.4. Extracellular matrix (ECM) score

ECM-related gene list contained 239 genes in three list (proteoglycans, collagens and glycoproteins) [35]. The expression levels of ECM-related genes were assessed using ssGSEA method in R package

GSVA v1.50.0 (https://github.com/rcastelo/GSVA).

2.5. 4D label-free mass spectrometry

Peptides from the primary skin fibroblasts were separated using EASY-nLC 1200 and analyzed via Orbitrap Exploris™ 480 according to the manufacturer's instructions. Raw files were processed using the MaxQuant search engine v.1.6.15.0 (FDR <1 %) (https://maxquant.org/maxquant/). Perseus v2.0.7.0 (https://maxquant.org/perseus/) was employed for the differential expression analysis of LFQ intensity. Only proteins that were detected in more than 70 % of samples were considered valid and included in subsequent normalization and paired testing.

2.6. Western blot

Equal amounts of protein isolated from primary skin fibroblast samples stimulated by TGF-β (4 controls and 4 TGF-β stimulated samples) at passage 6 were separated by 10 % SDS−PAGE and subsequently transferred to polyvinylidene difluoride membranes (PVDF). The blots were blocked with 5 % nonfat milk for 1 h, incubated at 4 °C overnight with *CTTN* antibody (Proteintech) and *GAPDH* antibody (Cell Signaling Technology), and then incubated at room temperature for 1 h with horseradish peroxidase-conjugated secondary antibody (Abiowell, China). The bands were enhanced using an enhanced chemiluminesce kit (NcmECL Ultra, New Cell & Molecular Biotech) and quantified by MiniChemi (SageCreation).

2.7. Intron and splicing factor network construction

After elimination of duplicates, 174 genes related to splicing pathways and spliceosomes were obtained from the PathCards database (https://pathcards.genecards.org/). The LASSO method and a Monte Carlo approach were employed to construct the Splicing Pathway-based Intron RegulatiOn Network network (SPIRONs) [23]. For each intron, 80 % of samples were randomly selected for the training set, and the rest were used as the test set. We built a LASSO model using penalty factor λ optimized by 3-fold cross-validation separately for patients and control, recording the predicted performance (R^2) and the regression coefficient (β) of each splicing factor. These steps were repeated 50 times to obtain an average R^2 and β. Only if the average R^2 was larger than 0.5 and the average absolute β was larger than 0.2, the intron and associated splicing factor will be selected to construct SPIRONs. Gephi v0.10.1 (https://gephi.org/) was employed to visualize the network.

2.8. Statistical analysis

Data analysis was conducted using R software v4.3.0 (https://www.r-project.org/). Following the Shapiro-Wilks test, Student's *t*-test or paired *t*-test were used to assess the statistical significance of differences for independent samples and paired samples respectively provided the data exhibited normal distribution with homogeneous variances, otherwise Mann-Whitney test was utilized. Similarly, correlations between the transcriptome and clinical characteristics were evaluated by Spearman correlation tests for not normally distributed data and Pearson correlation tests for normally distributed data with homogeneous variances. Differences were considered significant at P value < 0.05.

2.9. Weighted gene Co-expression network analysis network construction (WGCNA)

Totally 1954 differently expressed IRs (P value < 0.05) in WNT5a/JNK-i/ROCK-i/Actin-i/TGFβRI-i/ITGAV-i group were selected for WGCNA with the soft threshold determined by pickSoftThreshold function. Each module was set to have 30 IRs in minimum and all ungrouped IRs were classified to grey module. Gene significance (GS) and module membership (MM) were employed to assess the correlation among IRs, modules and phenotypes. Setting MM > 0.75 and GS > 0.75 as thresholds, IRs were selected to explore clinical associations.

3. Results

3.1. The alteration of AS events and clinical correlations in SSc

We initially conducted an analysis of AS events using bulk RNA-seq data obtained from the skin tissues of 48 SSc patients and 33 healthy controls (PRESS cohort) [32,36]. In all 81 skin samples, we identified a total of 236,153 AS events, comprising of skipping exon (SE, 71.04 %), alternative 3′ splice site (A3SS, 5.98 %), alternative 5′ splice site (A5SS, 4.06 %), mutually exclusive exons (MXE, 15.39 %), and intron retention (IR, 3.53 %). When compared with control, we identified significant AS events (FDR <0.05) in SSc. In these significant AS events, the proportion of SE decreased to 50.48 %, while the other four forms increased, with the most substantial changes observed in IR (12.28 %) and A5SS (6.48 %), which increased 247.93 % and 59.72 % respectively (Fig. 1A). The distribution of significant AS events exhibited variation across different chromosomes and forms (Fig. 1B). Specifically, SE events were predominant in chr1, chr2, and chr3, while MXE were predominantly found in chr1, chr2, and chr11. Chr1 showcased the highest number of significant AS events across all five forms (167 significant AS events). AS events (0.05 < Inclevel <0.95) were notably more prevalent in patients with a shorter disease duration (0–1.5 years) and higher local skin scores (score 2–3) compared with that in control, which increased 26.5 % and 21.1 % respectively. Noteworthy, patients without any SSc autoantibodies exhibited the highest number of AS events in comparison with that in control and patients with specific antibodies, namely anti-Topoisomerase antibody (ATA) and anti-RNA Polymerase III antibody (ARA) (Fig. 1C). Additionally, we identified a positive correlation between AS events and the extracellular matrix (ECM) score, proteoglycans score, collagens score, and glycoproteins score (Fig. 1D) [35]. Among 5 types of AS, the number of IR and A3SS events showed significantly positive correlation with ECM score, proteoglycans score, collagens score, collagens score and glycoproteins score (Fig. 1D). Furthermore, 1005 genes with significant AS events enriched pathways related to the extracellular matrix, including extracellular matrix organization and cell-substrate junction organization (Supplementary Fig. S1).

3.2. IR profiles in SSc skin tissue and clinical associations

As the most substantial changes of AS in SSc skin tissues is IR, we further analyzed the IR profiles in the skin tissues of PRESS cohort by our previously published software iREAD and GTFtools [19,20,32,36]. 8431 IRs originating from 2676 unique genes were identified. Among these, 95.8 % (8081/8431) IRs were derived from protein-coding genes and 2.95 % (249/8431) were from lncRNAs (Fig. 2A). IRs were 64.90 % shorter in length than non-IRs (Fig. 2B). Almost all of the introns from the same gene showed the same expression tendencies. We observed 2659 differentially expressed introns (DEIs) with P value < 0.05, of which 261 DEIs (24 downregulated and 237 upregulated) had an absolute log2-fold-change (log2FC) > 1 (Fig. 2C). A heatmap was generated showing the top 25 upregulated and 25 downregulated DEIs between patients and control, including genes associated with collagen synthesis (*COL11A1, COL8A1*) and genes from the cytochrome P450 family (*CYP4B1, CYP4Z2P, CYP4A11*) (Fig. 2D). Gene Ontology (GO) analysis of the 2659 DEIs revealed pathways associated with fibrosis, including wound healing, the Wnt signaling pathway, and collagen metabolic processes (Fig. 2E). Of the 2659 DEIs, 352 DEIs with no significantly different exons were enriched in pathways including cell growth, wound healing and ATP metabolic process (Supplementary Fig. S2).

IR events were significantly 10.16 % lower in patients with long

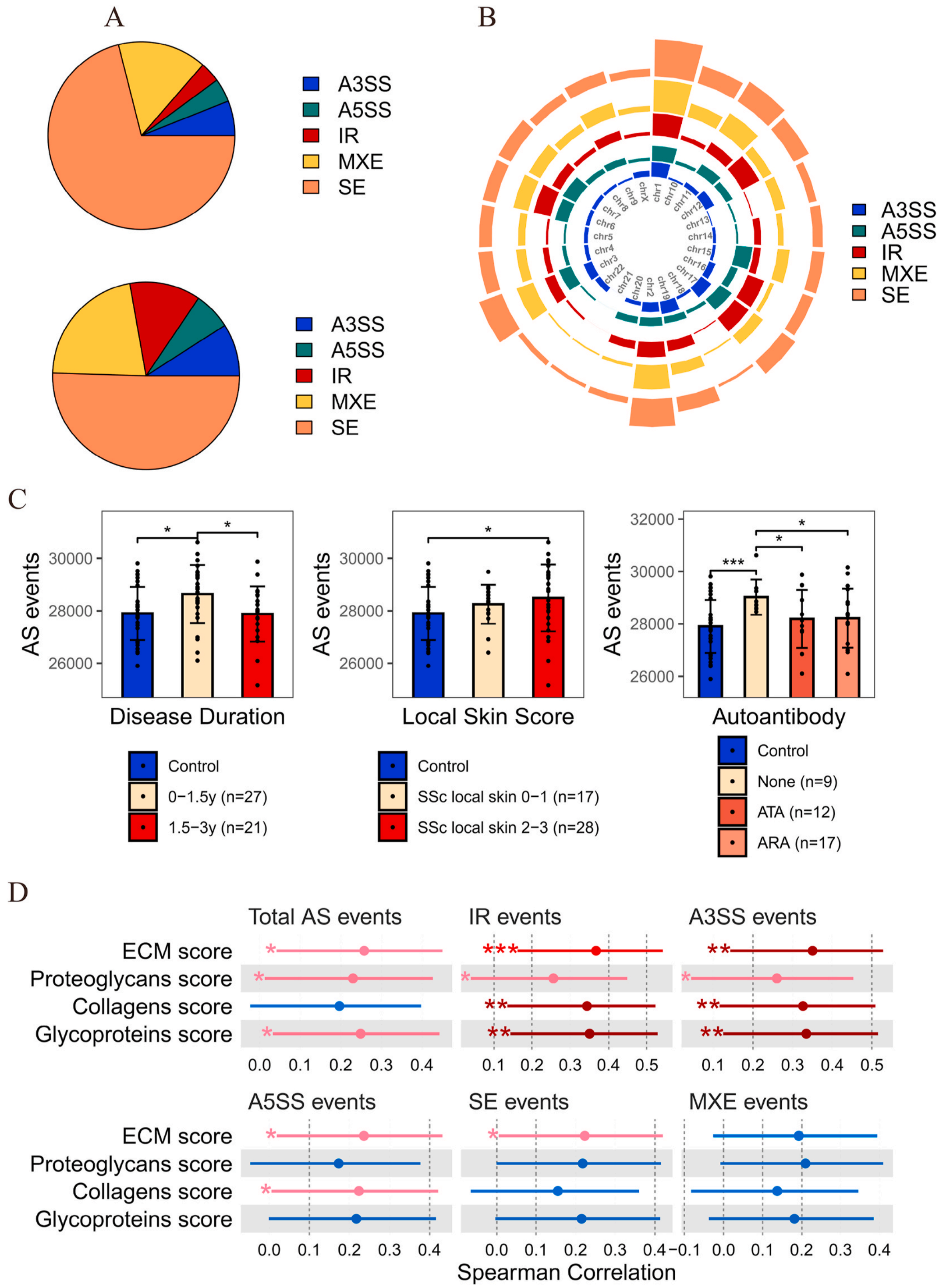

Fig. 1. Alternative splicing (AS) events in SSc skin tissues. (A) The upper panel illustrates the proportion of all identified AS events in the skin tissues of SSc and control, while the lower panel depicts the proportion of significant AS events (FDR<0.05) in SSc compared to control. (B) The distribution of significant AS events across chromosomes. (C) AS events in different clinical groups (*t*-test). (D) The correlation among AS events, IR events, A3SS events, A5SS events, MXE events and ECM scores. (*indicates P value < 0.05, **indicates P value < 0.01, ***indicates P value < 0.001, **** indicates P value < 0.0001).

disease duration (1.5–3 years) and 7.29 % lower in patients with interstitial lung disease (ILD) compared with control (when fibrosis occurred in lung, the symptoms were diagnosed as ILD. Nearly 40 % SSc patients are complicated by ILD [2]). Meanwhile, the IR events were significantly 7 % lower in SSc patients regardless of their modified Rodnan skin score (mRSS) and local skin score compared with control (Fig. 3A) (mRSS and local skin score were commonly used methods for assessing the skin thickness in SSc [2]). We identified DEIs in ECM-related genes and presented the top 20 genes with most DEIs (Fig. 3B). Furthermore, we explored the correlation between the expression of DEIs and the clinical features of SSc patients. As showed in Fig. 3C, among the top 25 upregulated and the 25 downregulated DEIs, *WIF1, THBS1, PLIN1, CYP2W1, COL8A1, and CFD* showed a significant correlation with total mRSS and local skin score. All these top DEIs were

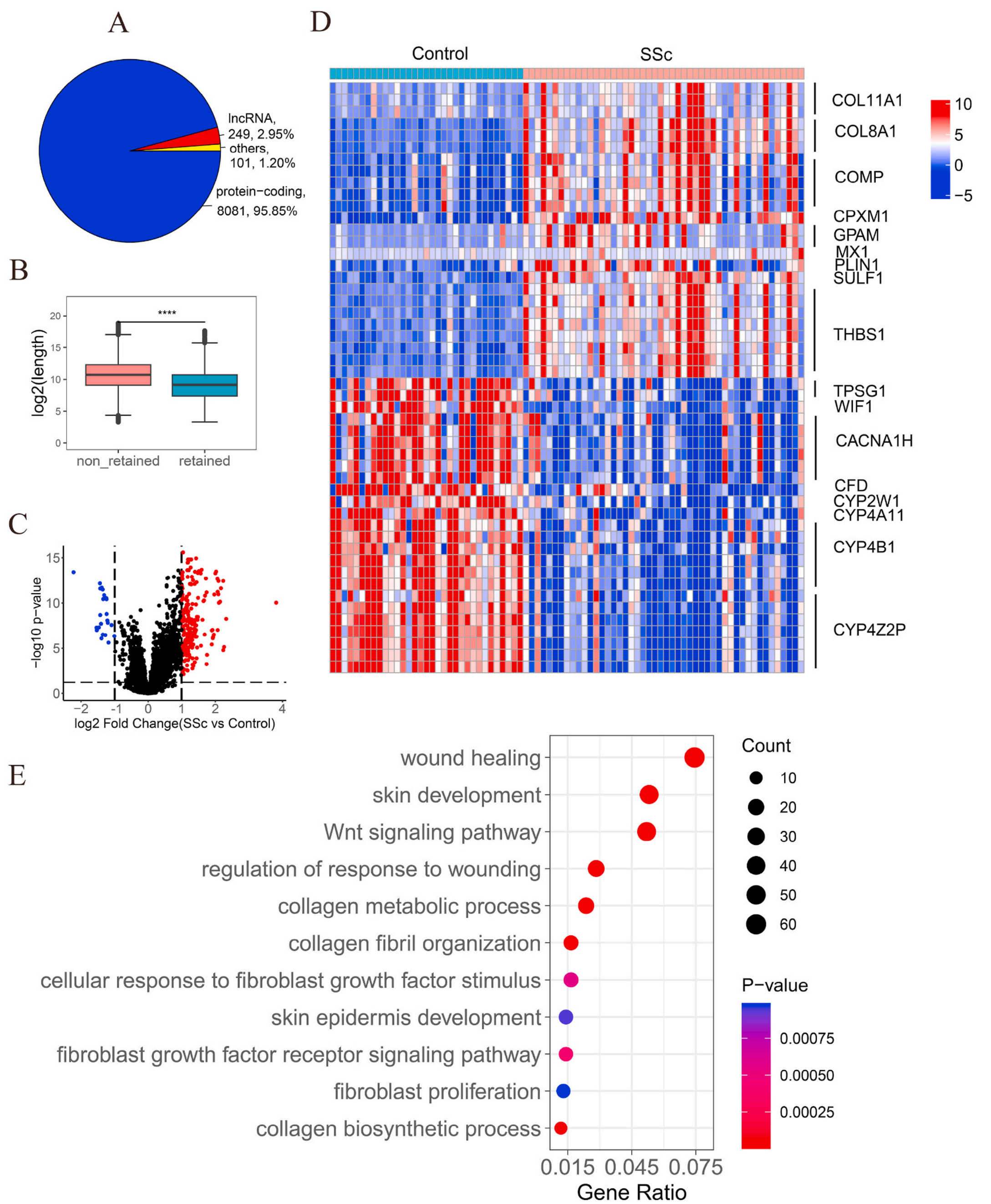

Fig. 2. Intron retention (IR) profiles in SSc skin tissues. (A) The genotype distribution of genes with IRs. (B) Length comparison between IRs and non-IRs (Mann-Whitney test). (C) Volcano plot displaying DEIs with P value < 0.05 and absolute log2FC > 1. (D) Heatmap based on the 50 most differentially expressed retained introns (DEIs) (top 25 upregulated and top 25 downregulated). (E) The fibrosis-related Gene Ontology terms enriched in all DEIs. (*indicates P value < 0.05, **indicates P value < 0.01, ***indicates P value < 0.001, **** indicates P value < 0.0001).

significantly associated with a previously established fibroblast signature score (Fibroblast) and histological parameters such as myofibroblast counts or relative collagen thickness [32].

3.3. *The influence of IR on protein expression in TGF-β-stimulated fibroblasts*

A total of 31282 AS events were identified in TGF-β stimulated fibroblasts, of which 2357 were significantly different from control. As the same with that of skin tissues, skipping exons occupied the largest proportion of AS events and IR had the largest change, increasing from 14.2 % in all AS events to 17.9 % in significant AS events (Supplementary Figs. S3A and B). Among 5 types of AS, the proportion of IR demonstrated positive correlation with extracellular matrix (ECM) score, collagens score, and glycoproteins score, while it presented negative correlation with proteoglycans score (Supplementary Fig. S3C). Genes with differently expressed AS events enriched fibrosis-related pathways, such as wound healing and fibroblast migration (Supplementary Fig. S3D).

In TGF-β-stimulated fibroblasts, we identified 3423 IR events, among which 93.6 % (3205/3423) of the introns were from protein-coding genes and 4.12 % (141/3423) belonged to lncRNAs (Fig. 4A). IRs were 81.93 % shorter than non-IRs (Fig. 4B). Differential expression analysis indicated that 9.93 % (340/3423) of the IRs differed significantly between stimulated fibroblasts and controls, with 187 downregulated and 142 upregulated DEIs (absolute log2FC greater than 1 and

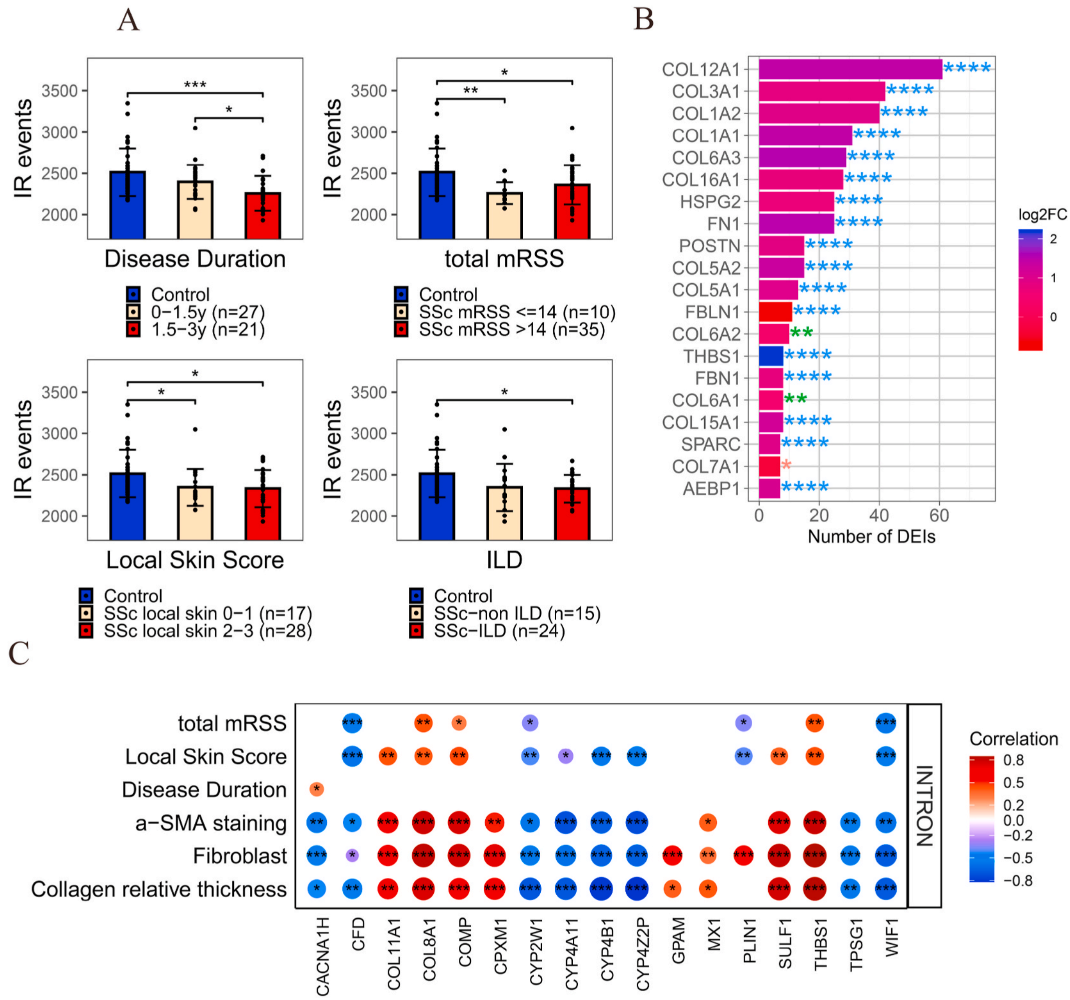

Fig. 3. The correlation of IR and clinical characteristics. (A) IR events in different clinical groups (Mann-Whitney test). (B) The top 20 ECM-related genes with most DEIs and their largest log2FC of DEI from the same gene. (C) The correlations between the top 50 DEIs and clinical characteristics. The size and color of the bubbles indicated the correlation coefficient. (*indicates P value < 0.05, **indicates P value < 0.01, ***indicates P value < 0.001). ILD: interstitial lung disease; total mRSS: total modified Rodnan skin score.

p value < 0.05; Fig. 4C). Among the top 25 upregulated and 25 downregulated DEIs, there were genes related to the cytoskeleton (*TPM1*), extracellular matrix (*COMP*), and TGF-β-induced genes (*PMEPA1, ADAM33*) (Fig. 4D). Pathway analysis suggested that the DEIs were involved in wound healing, response to TGF-β, collagen metabolic process, and fibroblast proliferation (Fig. 4E).

To study the influence of IR on protein expression, we performed a genome-wide analysis of intron, exon and protein expression. Genes with higher levels of IR tended to have lower levels of protein expression than genes with lower levels of IR (Fig. 5A). For example, when IR expression was elevated in TGF-β stimulated fibroblasts (log2FC-INTRON >0), the log2FC-protein value was consistently smaller than log2FC-INTRON, positioning the point below the dashed line (the dashed line merely indicated a diagonal trend). It indicated a negative correlation between IR expression and protein expression. To further clarify this relationship, a correlation analysis was conducted between protein and intron/exon ratio, taking into account the influence of exons. After correcting the expression of exons, the expression of introns was observed a significantly negative correlation with the expression of proteins both in stimulated fibroblasts and controls, which demonstrated that in cases where genes exhibit identical exon expression levels, elevated intron expression levels were associated with diminished protein expression (Fig. 5B).

Through a comprehensive analysis of introns, exons and proteins, we obtained 3 genes with upregulated IR, upregulated protein and no significant difference in exon expression (*CHPF, RPS8, FLNA*), 4 genes with downregulated IR, downregulated protein and no significant difference in exon expression (*ALDH1A3, ACADS, FAM180A, PHYKPL*), 8 genes with downregulated IR, upregulated protein and no significant difference in exon expression (*CTTN, OGA, HGH1, PPIB, LARP1, PRKCSH, MED16, RPL18A*), and 1 gene with downregulated IR and exon but upregulated protein (*TRIP6*) (Fig. 5C and Supplementary Fig. S4). We validated the expression levels of these sequences derived from IR and exons by RT−PCR and discovered that the expression levels of 4 genes (*CTTN, OGA, MED16, PHYKPL*) were consistent with the RNA-seq data (Fig. 5C). Western blot analysis showed that the significant differences in *CTTN* expression were consistent with the mass spectrometry results (Fig. 5D).

3.4. Impaired splicing machinery in SSc skin fibrosis

To understand how IR is regulated, we explored the correlation between IR and splicing factors. Using the LASSO method, we built Splicing Pathway-based Intron Regulation Network network (SPIRONs) for SSc patients and control separately using skin RNA-seq data (Supplementary Fig. S5). There was a significant difference between the SPIRONs of SSc patients and control. The SSc SPIRON demonstrated a well-organized structure comprising main splicing factors such as

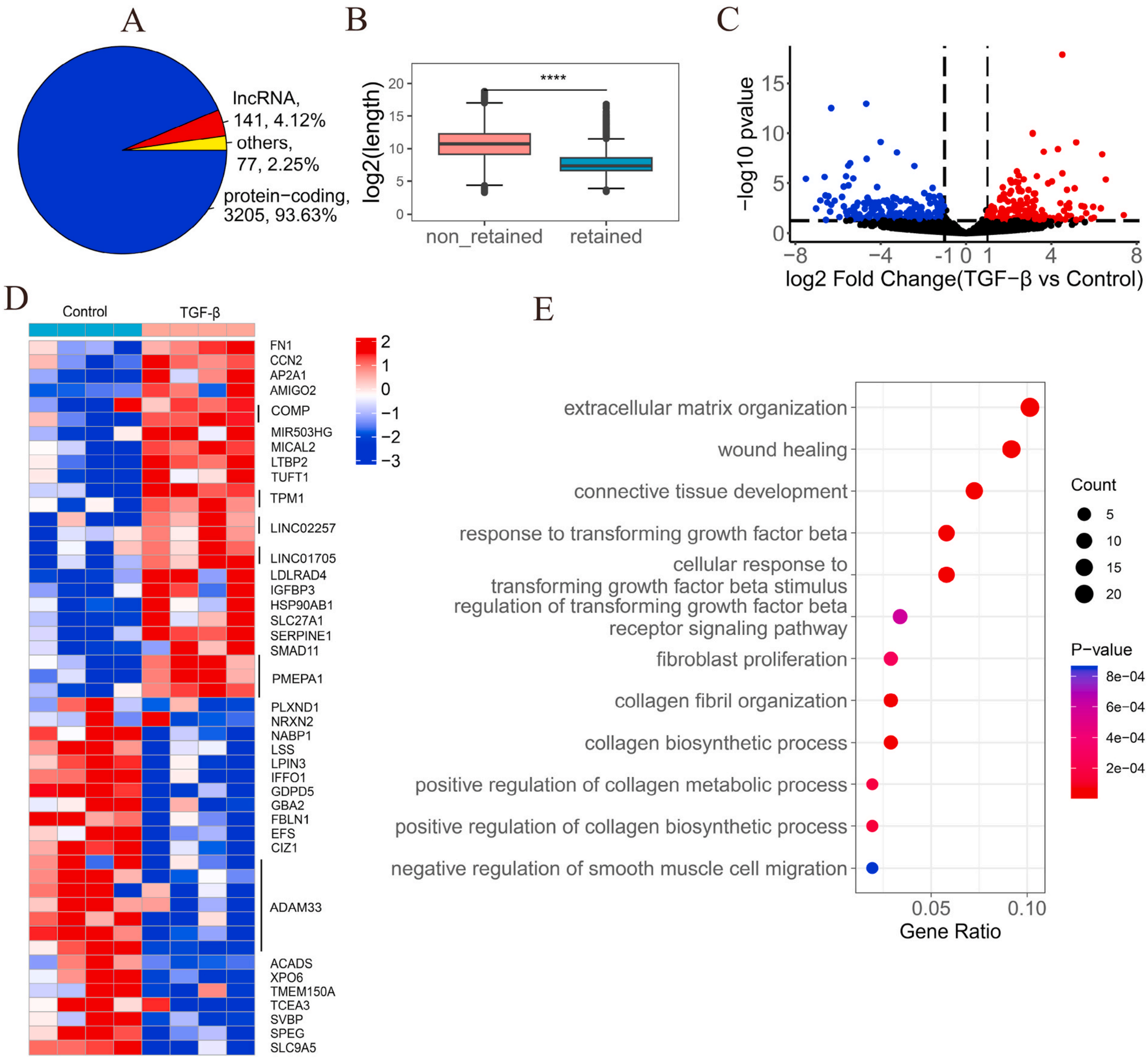

Fig. 4. Changes in IR in primary skin fibroblasts upon TGF-β stimulation. (A) The genotype distribution of genes with IRs. (B) Length comparison between IRs and non-IRs (Mann-Whitney test). (C) Volcano plot displaying 187 downregulated and 142 upregulated DEIs with absolute log2FC > 1 and P value < 0.05. (D) Heatmap of the 50 most differentially expressed DEIs (top 25 upregulated and top 25 downregulated). (E) The fibrosis-related Gene Ontology terms enriched in all DEIs (P value < 0.05).

POLR2F, NRPD3, SF3B3, DHX16, CRNKL1, and *RBMXL1*. Notably, we observed that some splicing factors (*SF3B3, CHERP*) appeared in both the SSc and control SPIRONs.

We screened for differentially expressed splicing factors and identified corresponding IRs from the SPIRONs. There were 36 downregulated splicing factors (log2FC < 0 and FDR<0.05) and 7 upregulated splicing factors (log2FC > 0 and FDR<0.05). As shown in Fig. 6A, upregulated *COLEC12* regulated IRs, such as *DSC, DMKN, PI16, DCN* in control-specific SPIRON, and *COL6A2, UBA7* in SSc-specific SPIRON. In SSc-specific SPIRON, collagen related genes (*COL1A1, COL1A2, and COL6A1*) occupied the most regulation by differentially expressed splicing factors. Fibrosis-related pathways, involving in extracellular matrix components and intercellular connections, were enriched by IRs regulated by differential splicing factors (Fig. 6B).

We further analyzed the clinical correlations of the 43 differentially expressed splicing factors. As showed in Fig. 6C, these splicing factors showed significant corrections with disease duration, local skin score, mRSS, fibroblast signature score (Fibroblast), and α-SMA staining score. Local skin score related the most splicing factors with the number 14. The highest correlation coefficient appeared in disease duration with *PQBP1* (Rho = 0.41, P = 0.0036) and α-SMA staining score with *DHX15* (Rho = 0.74, P = 0.0038).

3.5. IRs alteration rescued by treatment

To further validate whether IRs have the potential to be biomarkers of SSc, we conducted a three-step analysis in the revised manuscript. First, we compared the IR profiles in the baseline (n = 8) and follow-up (n = 10, with 2 patients undergoing three biopsies) skin tissues from the PRESS cohort to examine IR alterations before and after treatment. Next, we cultured fibroblasts in vitro and treated the cells with WNT5a and various inhibitors, including a JNK inhibitor, ROCK inhibitor, actin inhibitor, TGFβRI inhibitor, and ITGAV inhibitor. As shown in our previous study, WNT5a can activate TGF-β and facilitate fibroblast activation through the WNT5a/JNK/ROCK signaling pathway and integrin αV (*ITGAV*) [37]. Therefore, we examined the changes in IRs upon stimulation with WNT5a and the sequenced inhibitors. Finally, we analyzed fibroblasts exposed to various stimulations using GEO datasets, such as LiCl (GSE254716), JQ1 (BET inhibitor) (GSE186961), and TGF-β activated kinase 1 (*TAK1*) inhibitor (GSE232435).

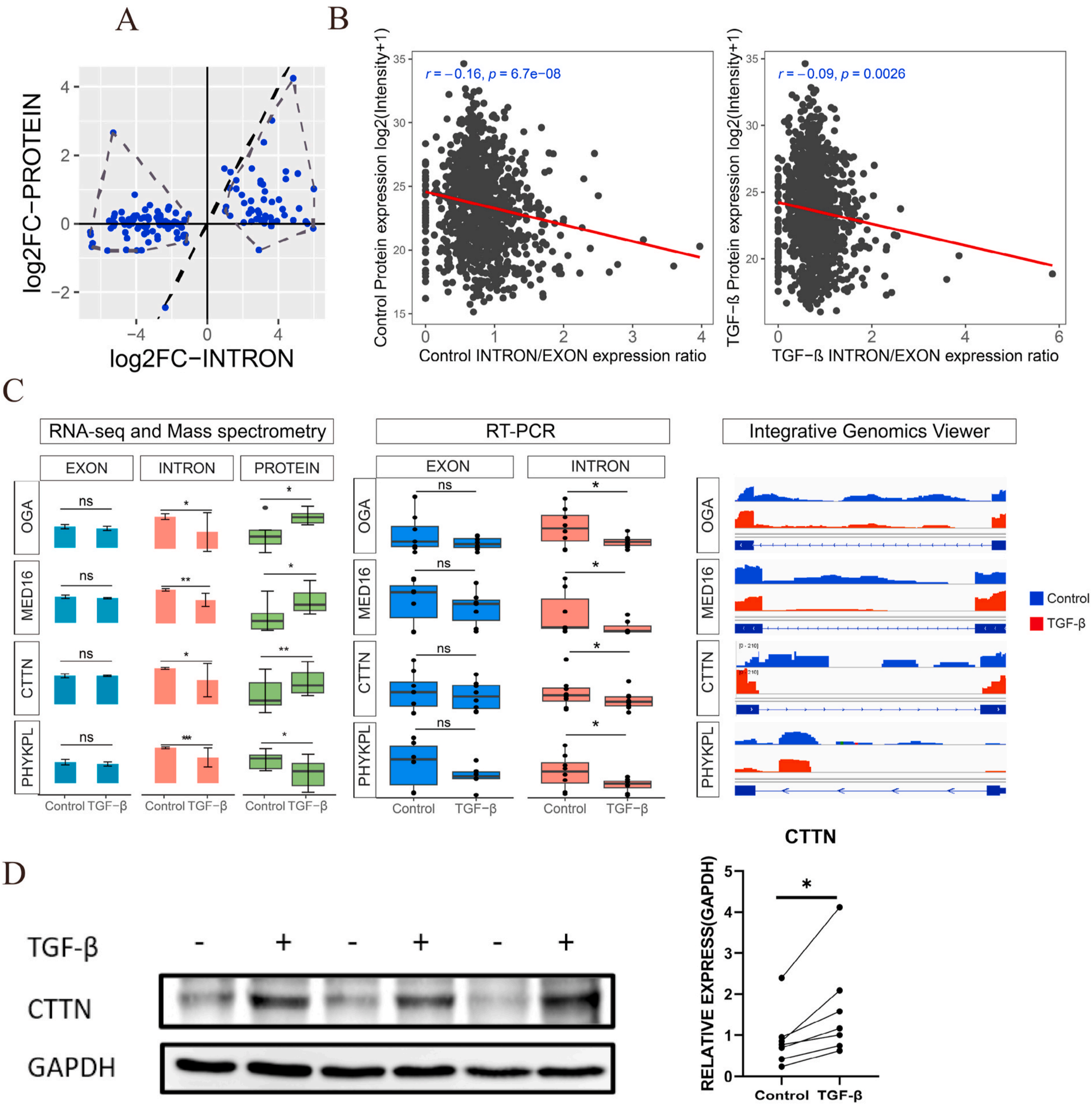

Fig. 5. The influences of IR on protein levels. (A) Genes with higher levels of intron expression tended to exhibit lower levels of protein expression than genes with lower levels of intron expression (the dashed line indicates the diagonal trend and was not the result of the linear regression model). As the figure shows, the log2FC-protein of genes with higher log2FC-intron tended to be closer to the x-axis compared with that of genes with lower log2FC-intron. When log2FC-intron>0, the points were located below the dashed line, and the points were located above the dashed line when the log2FC-intron<0. Taking the dashed line as the origin of these points, the points with higher absolute log2FC-intron values were vertically much farther from the dashed line compared with points with lower absolute log2FC-intron values. Some points did not follow this tendency, indicating that some IRs may evade NMD via other unknown mechanisms. (B) The spearman correlation between IR/exon ratio and protein expression in TGF-β-stimulated fibroblasts and controls. (C) Left: gene and protein expression levels of *CTTN, MED16, OGA, PHYKPL* detected by RNA-seq, mass spectrometer; Middle: gene expression of *CTTN, MED16, OGA, PHYKPL* validated by RT−PCR; Right: IR expression levels of *CTTN, MED16, OGA, PHYKPL* visualized by Integrative Genomics Viewer. (P values of intron/exon expression were calculated based on the negative binomial distribution in Deseq2 and paired *t*-test was used in the protein expression. In the results of RT-PCR, paired *t*-test was used in *PHYKPL* and paired Mann-Whitney test was used in *CTTN, OGA, MED16*. Paired Mann-Whitney test was employed in the WB result of *CTTN*) (D) Protein expression levels of *CTTN* detected by Western blotting and quantification of the results (*indicates P value < 0.05, **indicates P value < 0.01, ***indicates P value < 0.001, ****indicates P value < 0.0001, ns indicates P value > 0.05).

Compared to healthy controls, IR expression altered in immune-related genes in SSc skin tissues, with this alteration being modulated by treatment for over 6 months (mycophenolate mofetil/methotrexate), such as *CD44, CD74, IFI16* and *HLA-C* (Supplementary Fig. S6A). Additionally, some fibrosis-related genes presented IR alterations, such as *COL4A1*. The top 10 IRs with the greatest changes were selected for investigation of their relationships with clinical features. As Supplementary Fig. S6B showed, IRs in *C3* and *STAT1* exhibited positive correlation with a previously established fibroblast signature score (Fibroblast) [32], while IRs in *CD74, PTPRC, TLN1* and *XAF1* presented negative correlations with disease duration and total mRSS.

The IR profiles exhibited distinctive patterns within each group in response to WNT5a and various inhibitors, including the JNK inhibitor, ROCK inhibitor, actin inhibitor, TGFβRI inhibitor, and ITGAV inhibitor (Supplementary Fig. S7A). To further explore these changing patterns, we conducted weighted Gene Co-Expression Network Analysis (WGCNA)

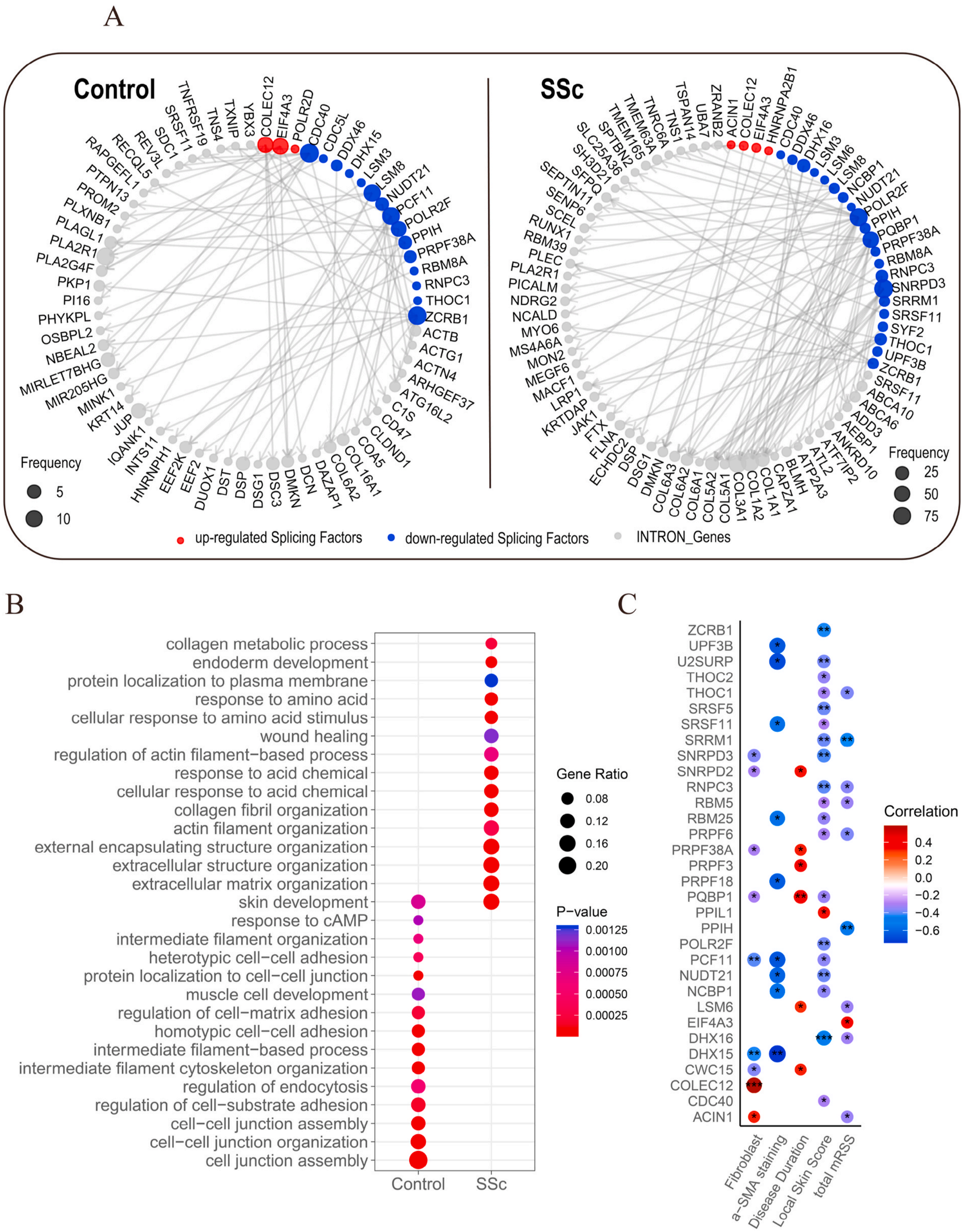

Fig. 6. Functions and clinical associations of altered splicing factors in SSc. (A) Differential splicing factors and their corresponding IRs in SPIRONs. (B) Pathways enriched in IRs regulated by differential splicing factors in skin tissue of SSc. (C) Spearman correlation between differential splicing factors and clinical features of SSc. The size and color of the bubbles indicated the correlation coefficient. (*indicates P value < 0.05, **indicates P value < 0.01, ***indicates P value < 0.001).

based on the differentially expressed IRs in each group (1954 IRs in total). This analysis divided the genes into different modules according to their alteration patterns in the different groups. Our analysis identified 15 distinct IR changing patterns, with the brown module (274 IRs) presenting the highest levels in WNT5a and decreasing in all groups treated with JNK/ROCK/TGFβRI/Actin/ITGAV inhibitors (Supplementary Figs. S7B and C). The top 11 changed IRs, derived from 5 extracellular matrix-related genes, showed clinical associations, including disease duration, local skin score, and mRSS. Six IRs of *COL1A1* demonstrated a positive correlation with local skin score, fibroblast signature, collagen staining score, and α-SMA staining score, while only *COL1A1*_35 showed a negative correlation with disease duration (Supplementary Fig. S7D).

Meanwhile, we identified alterations in IR profiles in fibroblasts exposed to various inhibitors, including LiCl (GSE254716), JQ1 (BET inhibitor) (GSE186961), and TGF-β activated kinase 1 (*TAK1*) inhibitor (GSE232435). Compared to untreated SSc fibroblasts, some differentially expressed IRs in LiCl/JQ1-treated fibroblasts originated from fibrosis-related genes (Supplementary Figs. S8A and B). For example, *COL1A1*, which exhibited a considerable number of up-regulated IRs in SSc skin tissues compared to controls, showed down-regulated IRs in LiCl/JQ1-treated SSc fibroblasts compared to untreated SSc fibroblasts. In Supplementary Fig. S8C, multiple IRs were up-regulated by TGF-β stimulation and rescued by the TAK1 inhibitor, including fibrosis-related genes such as *COL1A1, ELN, POSTN, and FN1*.

4. Discussion

Our research unveiled noteworthy changes in AS events in the SSc skin tissues compared with that in control. The most significant alterations in AS forms were observed in IR and SE. Furthermore, AS events were identified in numerous ECM-related genes and demonstrated correlations with clinical indices and ECM scores. Consequently, AS may be implicated in the pathogenesis of SSc by enhancing transcriptomic diversity to increase protein isoforms or regulate protein abundances.

Further analysis indicated that IR profiles were modified not only in SSc skin tissues but also in fibroblasts stimulated with TGF-β. Genetic polymorphism of *CYP2D6* was reported to be associated with incidence of SSc and *CYP1A1* was downregulated in fibroblasts of SSc [38,39]. We noticed that genes from the cytochrome P450 family exhibited downregulated DEIs and downregulated exon, suggesting a potential defect on metabolism in SSc patients. Several IRs exhibited correlations with clinical features. Certain genes, marked by DEIs and nonsignificant exons, emerged as promising candidates for novel biomarkers [35, 36]. For instance, the IR of *JAK1*, a target of small molecule inhibitors, correlated with the mRSS and local skin score, while its exon expression did not correlate with any of the investigated features. This suggested that previous transcriptome sequencing focused on exon expression may have ignored some important information, while IR may provide additional information.

DEIs observed in SSc skin tissues and TGF-β-stimulated fibroblasts were found to actively participate in fibrosis-related pathways, including wound healing, the Wnt signaling pathway, and fibroblast proliferation. Substantial changes in IR events were also found in fibroblasts upon TGF-β stimulation. All these findings underscore the crucial role of alterations in IR profiles in the context of SSc fibrosis.

IR derived from different gene types exhibited disparate functional characteristics. IRs in lncRNA were observed to prompt nuclear localization while IR in protein-coding genes was found to be involved in translation processes mainly [40]. A previous study demonstrated IRs were widespread occurring in nearly 88 % protein-coding genes across all tissues and elevated IR expression levels were involved in reduced protein levels [41]. The mechanisms and effects of IR on proteins are diverse [41,42]. First, IRT stalling in the nucleus and degradation of IRTs in the cytoplasm via nonsense mediated decay (NMD) reduce translation [16,43]. Introns containing premature termination codons (PTCs) end translation inappropriately, leading to the generation of truncated proteins that do not function correctly or even accumulate and cause damage to normal cell components [26,44–46]. In addition, introns can bind to miRNAs, inducing translation inhibition [8]. To explore the function of IR on proteins, we collected protein from TGF-β-stimulated fibroblasts. For the majority of genes, the protein expression tended to decrease with a higher level of IR. The IR/exon ratio demonstrated a negative correlation with protein abundance, reflecting the impact of IRs on protein expression. Our data suggested that the changes in IR and protein levels may not align consistently. Additionally, Introns, with no significantly different exons, were observed to upregulate or downregulate protein expression. Through bioinformatic analysis, PTCs were found in the sequences of 16 introns that we identified from TGF-β-stimulated fibroblasts. Nine genes exhibited a distinct contrast in intron and protein expression tendencies, which could be attributed to NMD. However, it remains unclear whether the remaining genes evade NMD via unidentified mechanisms. Further experimentation is necessary to establish whether these introns regulate proteins directly and what mechanisms they employ.

The exact mechanisms of IR remain unclear. The sequence features of introns contribute partially to the process of IR [47]. Numerous researches proved that IRs were shorter than non-retained introns [23,48,49]. Shorter lengths make the splicing sites besides introns more challenging to be recognized and introns tend to resemble exons to a greater extent in order to be retained [49]. Low levels of methylation and a slow rate of elongation causes reduced occupancy of methyl-CpG binding protein 2 (MeCP2), resulting in decreased splicing factor recruitment, and subsequent IR alteration [50,51]. Additionally, we observed abnormal splicing factor expression in SSc with clinical associations and speculated that splicing factor dysfunction ultimately result in abnormal IR. The dysregulation of splicing factors has been validated as contributing to the pathogenesis of multiple diseases and specific drugs have been developed to target splicing factors [31,52]. Several splicing factors were found to be correlated with fibrosis, such as *SF3B3* and *CHERP*. *SF3B3* acts as a component of U2 snRNP, assisting U2 snRNP to anchor to pre-mRNA [53]. *CHERP* forms a part of U2 snRNP and contributes to calcium homeostasis [54]. Splicing factors that dominated the SPIRONs of control but were on the edges of the SPIRONs of SSc patients, such as *SNRPD3*, or that appeared exclusively in the SSc SPIRON network could have potential as new therapeutic targets in SSc. *SNRPD3*, a gene in the center of the SSc SPIRON network, is a core component of the spliceosome and an antigen of the anti-Smith antibody in SLE patients [7,55].

This research had some limitations. First, the exact functions of IR were not confirmed in animal models or in vitro experiments, and the role of IR in fibrotic pathways was not clarified. Next, the correlations between clinical indices and IR still need to be validated in more cohorts. Finally, splicing machinery dysfunction has not yet been confirmed as the regulatory factors of IR. The process from transcription to translation is complex and regulated by multiple factors, including DNA methylation, AS, miRNAs and so on. Our research only provided an insight into fibrosis mechanisms and attempted to find upstream regulators which have the potential to be targets of treatment.

In total, we identified abnormal AS and IR profiles, along with splicing factors, in SSc skin fibrosis. These aberrations were demonstrated to impact protein expression and play a role in fibrosis processes, exhibiting distinct distributions among fibroblast subtypes and in response to fibroblast activation.

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Ethics approval

The study involving human participants were reviewed and approved by the ethics committee of Xiangya Hospital (201303293).

Competing interests

None declared.

CRediT authorship contribution statement

Shasha Xie: Writing – original draft, Formal analysis, Conceptualization, Investigation. **Ding Bao:** Validation, Investigation, Writing – review & editing. **Yizhi Xiao:** Software, Methodology. **Hongdong Li:** Software, Methodology. **Muyao Guo:** Resources. **Bingying Dai:** Resources. **Sijia Liu:** Funding acquisition. **Jing Huang:** Resources. **Muyuan Li:** Resources. **Liqing Ding:** Resources. **Qiming Meng:** Resources. **Chun-Liu Lv:** Resources. **Jörg H.W. Distler:** Writing – review & editing, Resources. **Hui Luo:** Supervision, Funding acquisition. **Honglin Zhu:** Writing – review & editing, Funding acquisition, Conceptualization, Supervision.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jaut.2024.103306.

References

[1] F. van den Hoogen, D. Khanna, J. Fransen, S.R. Johnson, M. Baron, A. Tyndall, et al., Classification criteria for systemic sclerosis: an American college of rheumatology/European league against rheumatism collaborative initiative, Ann. Rheum. Dis. 2013 (72) (2013) 1747–1755.
[2] Y. Allanore, R. Simms, O. Distler, M. Trojanowska, J. Pope, C.P. Denton, et al., Systemic sclerosis, Nat. Rev. Dis. Prim. 1 (2015) 15002.
[3] Y.Y. Ho, D. Lagares, A.M. Tager, M. Kapoor, Fibrosis–a lethal component of systemic sclerosis, Nat. Rev. Rheumatol. 10 (2014) 390–402.
[4] C. Varrica, H.S. Dias, C. Reis, M. Carvalheiro, S. Simoes, Targeted delivery in scleroderma fibrosis, Autoimmun. Rev. 20 (2021) 102730.
[5] B. Schwanhausser, D. Busse, N. Li, G. Dittmar, J. Schuchhardt, J. Wolf, et al., Global quantification of mammalian gene expression control, Nature 473 (2011) 337–342.
[6] L.E. Marasco, A.R. Kornblihtt, The physiology of alternative splicing, Nat. Rev. Mol. Cell Biol. 24 (2023) 242–254.
[7] A. Hegele, A. Kamburov, A. Grossmann, C. Sourlis, S. Wowro, M. Weimann, et al., Dynamic protein-protein interaction wiring of the human spliceosome, Mol Cell 45 (2012) 567–580.
[8] U. Schmitz, N. Pinello, F. Jia, S. Alasmari, W. Ritchie, M.C. Keightley, et al., Intron retention enhances gene regulatory complexity in vertebrates, Genome Biol. 18 (2017) 216.
[9] J. Merkin, C. Russell, P. Chen, C.B. Burge, Evolutionary dynamics of gene and isoform regulation in mammalian tissues, Science 338 (2012) 1593–1599.
[10] H. Dvinge, R.K. Bradley, Widespread intron retention diversifies most cancer transcriptomes, Genome Med. 7 (2015) 45.
[11] J. Parenteau, L. Maignon, M. Berthoumieux, M. Catala, V. Gagnon, S. Abou Elela, Introns are mediators of cell response to starvation, Nature 565 (2019) 612–617.
[12] I.D. Green, N. Pinello, R. Song, Q. Lee, J.M. Halstead, C.T. Kwok, et al., Macrophage development and activation involve coordinated intron retention in key inflammatory regulators, Nucleic Acids Res. 48 (2020) 6513–6529.
[13] O. Mauger, F. Lemoine, P. Scheiffele, Targeted intron retention and excision for rapid gene regulation in response to neuronal activity, Neuron 92 (2016) 1266–1278.
[14] T. Ni, W. Yang, M. Han, Y. Zhang, T. Shen, H. Nie, et al., Global intron retention mediated gene regulation during CD4+ T cell activation, Nucleic Acids Res. 44 (2016) 6817–6829.
[15] C.R. Edwards, W. Ritchie, J.J. Wong, U. Schmitz, R. Middleton, X. An, et al., A dynamic intron retention program in the mammalian megakaryocyte and erythrocyte lineages, Blood 127 (2016) e24–e34.
[16] H. Pimentel, M. Parra, S.L. Gee, N. Mohandas, L. Pachter, J.G. Conboy, A dynamic intron retention program enriched in RNA processing genes regulates gene expression during terminal erythropoiesis, Nucleic Acids Res. 44 (2016) 838–851.
[17] E.T. Wang, R. Sandberg, S. Luo, I. Khrebtukova, L. Zhang, C. Mayr, et al., Alternative isoform regulation in human tissue transcriptomes, Nature 456 (2008) 470–476.
[18] D.L. Black, Protein diversity from alternative splicing: a challenge for bioinformatics and post-genome biology, Cell 103 (2000) 367–370.
[19] H.D. Li, C.X. Lin, J. Zheng Gtftools, A software package for analyzing various features of gene models, Bioinformatics 38 (2022) 4806–4808.
[20] H.D. Li, C.C. Funk, N.A.-O. Price, iREAD: a tool for intron retention detection from RNA-seq data, BMC Genom. 21 (2020) 128.
[21] N. Altemose, G.A. Logsdon, A.V. Bzikadze, P. Sidhwani, S.A. Langley, G.V. Caldas, et al., Complete genomic and epigenetic maps of human centromeres, Science 376 (2022) eabl4178.
[22] T. Kanholm, U. Rentia, M. Hadley, J.A. Karlow, O.L. Cox, N. Diab, et al., Oncogenic transformation drives DNA methylation loss and transcriptional activation at transposable element loci, Cancer Res. 83 (2023) 2584–2599.
[23] H.D. Li, C.C. Funk, K. McFarland, E.B. Dammer, M. Allen, M.M. Carrasquillo, et al., Integrative functional genomic analysis of intron retention in human and mouse brain with Alzheimer's disease, Alzheimers Dement 17 (2021) 984–1004.
[24] Y. Xiao, S. Xie, H.D. Li, Y. Liu, H. Zhang, X. Zuo, et al., Characterised intron retention profiles in muscle tissue of idiopathic inflammatory myopathy subtypes, Ann. Rheum. Dis. 83 (7) (2024) 901–914, https://doi.org/10.1136/ard-2023-225035.
[25] Y.C. Hsieh, C. Guo, H.K. Yalamanchili, M. Abreha, R. Al-Ouran, Y. Li, et al., Tau-mediated disruption of the spliceosome triggers cryptic RNA splicing and neurodegeneration in Alzheimer's disease, Cell Rep. 29 (2019) 301–316 e10.
[26] Z.K. Ngian, Y.Y. Tan, C.T. Choo, W.Q. Lin, C.Y. Leow, S.J. Mah, et al., Truncated Tau caused by intron retention is enriched in Alzheimer's disease cortex and exhibits altered biochemical properties, Proc Natl Acad Sci U S A 119 (2022) e2204179119.
[27] P.D. James, S. Raut, G.E. Rivard, M.C. Poon, M. Warner, S. McKenna, et al., Aminoglycoside suppression of nonsense mutations in severe hemophilia, Blood 106 (2005) 3043–3048.
[28] M. Seiler, A. Yoshimi, R. Darman, B. Chan, G. Keaney, M. Thomas, et al., H3B-8800, an orally available small-molecule splicing modulator, induces lethality in spliceosome-mutant cancers, Nat Med 24 (2018) 497–504.
[29] E.G. Folco, K.E. Coil, R. Reed, The anti-tumor drug E7107 reveals an essential role for SF3b in remodeling U2 snRNP to expose the branch point-binding region, Genes Dev. 25 (2011) 440–444.
[30] S.W. Peltz, M. Morsy, E.M. Welch, A. Jacobson, Ataluren as an agent for therapeutic nonsense suppression, Annu. Rev. Med. 64 (2013) 407–425.
[31] R.J. van Alphen, E.A. Wiemer, H. Burger, F.A. Eskens, The spliceosome as target for anticancer treatment, Br. J. Cancer 100 (2009) 228–232.
[32] B.A.-O. Skaug, D.A.-O. Khanna, W.R. Swindell, M.E. Hinchcliff, T.M. Frech, V. D. Steen, et al., Global skin gene expression analysis of early diffuse cutaneous systemic sclerosis shows a prominent innate and adaptive inflammatory profile, Ann. Rheum. Dis. 79 (2020) 379–386.
[33] S. Shen, J.W. Park, Z.-x. Lu, L. Lin, M.D. Henry, Y.N. Wu, et al., rMATS: Robust and Flexible Detection of Differential Alternative Splicing from Replicate RNA-Seq Data, vol. 111, Proceedings of the National Academy of Sciences, 2014.
[34] M.D. Robinson, G.K. McCarthy Dj Fau - Smyth, G.K. Smyth, edgeR: a Bioconductor package for differential expression analysis of digital gene expression data, Bioinformatics 26 (2010) 139–140.
[35] C. Kuppe, M.M. Ibrahim, J. Kranz, X. Zhang, S. Ziegler, J. Perales-Patón, et al., Decoding myofibroblast origins in human kidney fibrosis, Nature 589 (2020) 281–286.
[36] S. Jaafar, A. Lescoat, S. Huang, J. Gordon, M. Hinchcliff, A.A. Shah, et al., Clinical characteristics, visceral involvement, and mortality in at-risk or early diffuse systemic sclerosis: a longitudinal analysis of an observational prospective multicenter US cohort, Arthritis Res. Ther. 23 (2021) 170.
[37] T. Trinh-Minh, C.-W. Chen, C. Tran Manh, Y.-N. Li, H. Zhu, X. Zhou, et al., Noncanonical WNT5A controls the activation of latent TGF-β to drive fibroblast activation and tissue fibrosis, J. Clin. Invest. 134 (2024).
[38] M. Barańska, B. Dziankowska-Bartkowiak, E. Waszczykowska, M. Rychlik-Sych, J. Skrętkowicz, Significance of genetic polymorphism of CYP2D6 in the pathogenesis of systemic sclerosis, Pharmacol. Rep. 64 (2012) 336–342.
[39] Y. Shi, B. Tang, J. Yu, Y. Luo, Y. Xiao, Z. Pi, et al., Aryl hydrocarbon receptor signaling activation in systemic sclerosis attenuates collagen production and is a potential antifibrotic target, Int Immunopharmacol 88 (2020) 106886.
[40] G. Dumbović, U. Braunschweig, H.K. Langner, M. Smallegan, J. Biayna, E.P. Hass, et al., Nuclear compartmentalization of TERT mRNA and TUG1 lncRNA is driven by intron retention, Nat. Commun. 12 (2021) 3308.
[41] R. Middleton, D. Gao, A. Thomas, B. Singh, A. Au, J.J. Wong, et al., IRFinder: assessing the impact of intron retention on mammalian gene expression, Genome Biol. 18 (2017) 51.
[42] Q. Wang, E.G. Conlon, J.L. Manley, D.C. Rio, Widespread intron retention impairs protein homeostasis in C9orf72 ALS brains, Genome Res. 30 (2020) 1705–1715.
[43] O.J. Ziff, D.M. Taha, H. Crerar, B.E. Clarke, A.M. Chakrabarti, G. Kelly, et al., Reactive astrocytes in ALS display diminished intron retention, Nucleic Acids Res. 49 (2021) 3168–3184.
[44] S. Tommasi, T.H. Kitapci, H. Blumenfeld, A. Besaratinia, Secondhand smoke affects reproductive functions by altering the mouse testis transcriptome, and leads to select intron retention in Pde1a, Environ. Int. 161 (2022) 107086.
[45] C.E. Comstock, M.A. Augello, R.P. Benito, J. Karch, T.H. Tran, F.E. Utama, et al., Cyclin D1 splice variants: polymorphism, risk, and isoform-specific regulation in prostate cancer, Clin. Cancer Res. 15 (2009) 5338–5349.
[46] A.C. Smart, C.A. Margolis, H. Pimentel, M.X. He, D. Miao, D. Adeegbe, et al., Intron retention is a source of neoepitopes in cancer, Nat. Biotechnol. 36 (2018) 1056–1058.
[47] U. Braunschweig, N.L. Barbosa-Morais, Q. Pan, E.N. Nachman, B. Alipanahi, T. Gonatopoulos-Pournatzis, et al., Widespread intron retention in mammals functionally tunes transcriptomes, Genome Res. 24 (2014) 1774–1786.
[48] S. Stamm, J. Zhu, K. Nakai, P. Stoilov, O. Stoss, M.Q. Zhang, An alternative-exon database and its statistical analysis, DNA Cell Biol. 19 (2000) 739–756.
[49] N.J. Sakabe, S.J. de Souza, Sequence features responsible for intron retention in human, BMC Genom. 8 (2007) 59.
[50] J.J. Wong, W. Ritchie, O.A. Ebner, M. Selbach, J.W. Wong, Y. Huang, et al., Orchestrated intron retention regulates normal granulocyte differentiation, Cell 154 (2013) 583–595.
[51] J.J. Wong, D. Gao, T.V. Nguyen, C.T. Kwok, M. van Geldermalsen, R. Middleton, et al., Intron retention is regulated by altered MeCP2-mediated splicing factor recruitment, Nat. Commun. 8 (2017) 15134.

[52] A. Ibanez-Costa, C. Perez-Sanchez, A.M. Patino-Trives, M. Luque-Tevar, P. Font, I. Arias de la Rosa, et al., Splicing machinery is impaired in rheumatoid arthritis, associated with disease activity and modulated by anti-TNF therapy, Ann. Rheum. Dis. 81 (2022) 56–67.
[53] C.L. Will, H. Urlaub, T. Achsel, M. Gentzel, M. Wilm, R. Lührmann, Characterization of novel SF3b and 17S U2 snRNP proteins, including a human Prp5p homologue and an SF3b DEAD-box protein, The EMBO journal 21 (2002) 4978–4988.
[54] A. De Maio, H.K. Yalamanchili, C.J. Adamski, V.A. Gennarino, Z. Liu, J. Qin, et al., RBM17 interacts with U2SURP and CHERP to regulate expression and splicing of RNA-processing proteins, Cell Rep. 25 (2018) 726–736 e7.
[55] S.O. Hoch, R.A. Eisenberg, G.C. Sharp, Diverse antibody recognition patterns of the multiple Sm-D antigen polypeptides, Clin. Immunol. 92 (1999) 203–208.